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MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, THERMOCOUPLE EHU-4/A

This specification is approved for use by HQ AFLC CASO/ LODS, Federal Center Battle Creek, MI 49016, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a single pointer, integrally lighted, 2-inch size case, thermocouple type temperature indicator, designated EHU-4/A.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Military

MIL-P-116	Preservation - Packaging, Methods of
MIL-I-7071	Indicator, Temperature, Thermocouple, Hermetically Sealed, General Specification For
MIL-L-25467	Lighting, Integral, Red, Aircraft Instrument, General Specification For
MIL-C-25516/17	Connector, Electrical, Miniature Coaxial Enviromental Resistant Type Coaxial Hermetically Sealed Receptacle, Type 25516/17

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking For Shipment and Storage

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: HQ AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016 by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6685

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Safety of individual instruments can be verified by analytical methods or by testing. Laboratory tests for

STANDARDS

Military

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS24367	Lamp - Incandescent, Miniature, Integral Lighting
MS33639	Case, Instrument, Clamp-Mounted, Aircraft
MS35214	Screw, Machine-Pan Head, Cross-Recessed, Brass, UNC-2A
MS35215	Screw, Machine-Pan Head, Cross-Recessed, Brass, Black Oxide, UNF-2A
MS35338	Washer, Lock-Spring, Helical Regular (Medium) Series

Air Force-Navy Aeronautical

AN961	Washer -Flat, Electrical

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The indicator furnished under this specification shall be a product which has been tested and has passed the qualification tests herein.

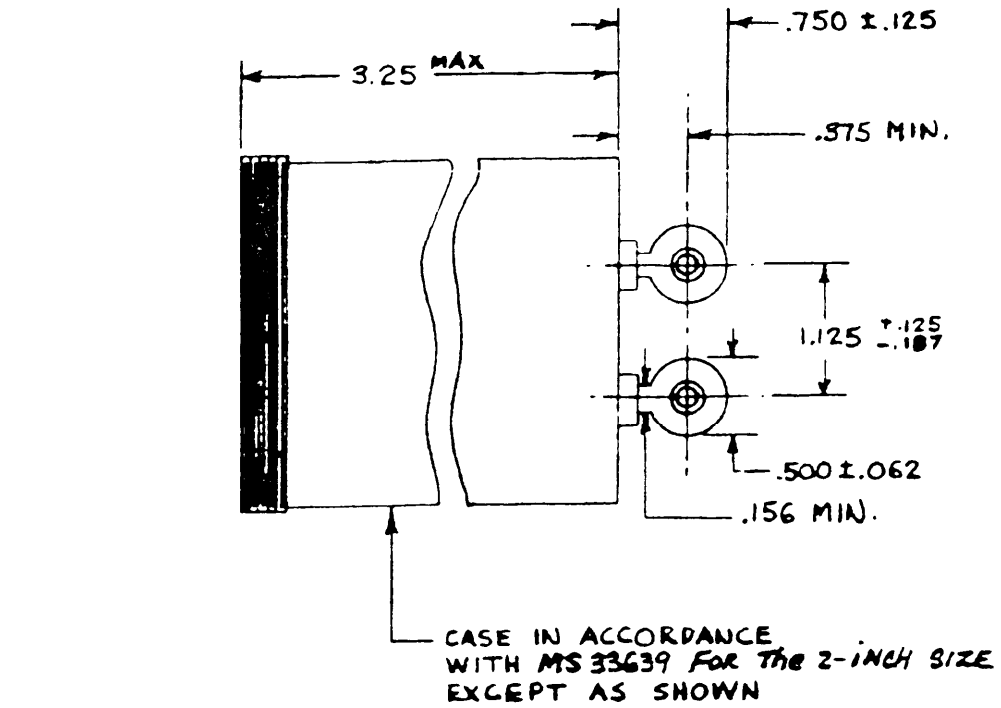
3.2 General specification. The requirements of MIL-I-7071 apply as requirements of this specification with the exceptions and additions called out herein. When the two specification conflict, this specification shall govern.

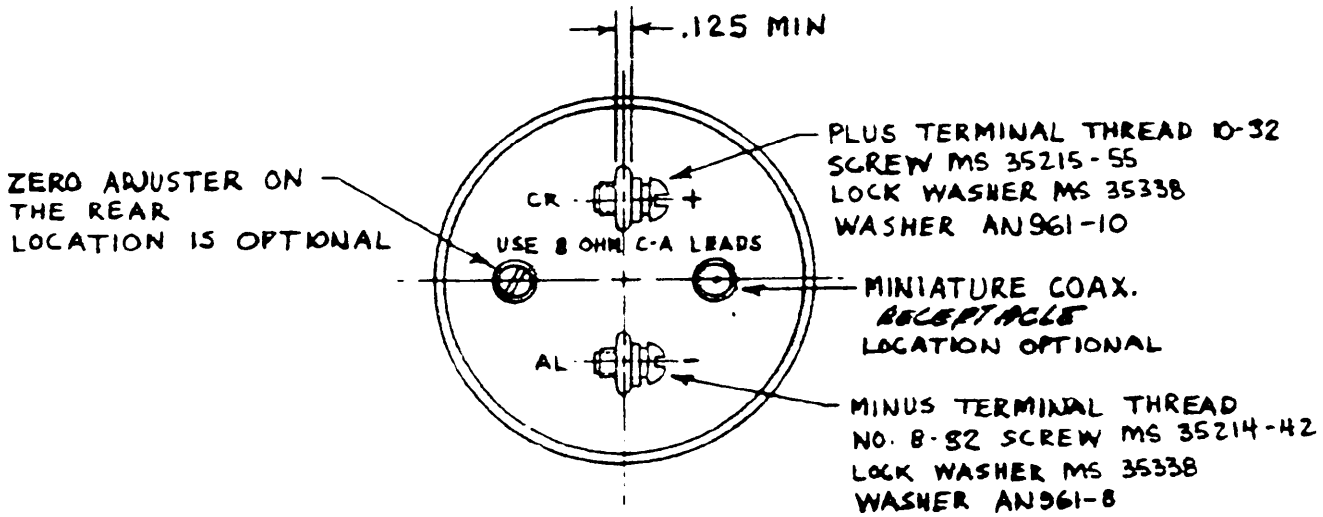
3.3 DESIGN AND CONSTRUCTION

3.3.1 General. The case detail of the indicator shall conform to figure 1.

*3.3.2 Dial. The dial of the indicator shall conform to figure 2. The 700°C scale graduation shall be located on the horizontal centerline, left side of the dial. The dial blank shall consist of a flat disc of plastic, metal, or

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REAR VIEW

BREAK SHARP EDGES APPROX. .010 RADIUS
DIMENSIONS IN INCHES

Figure 1. Case Details

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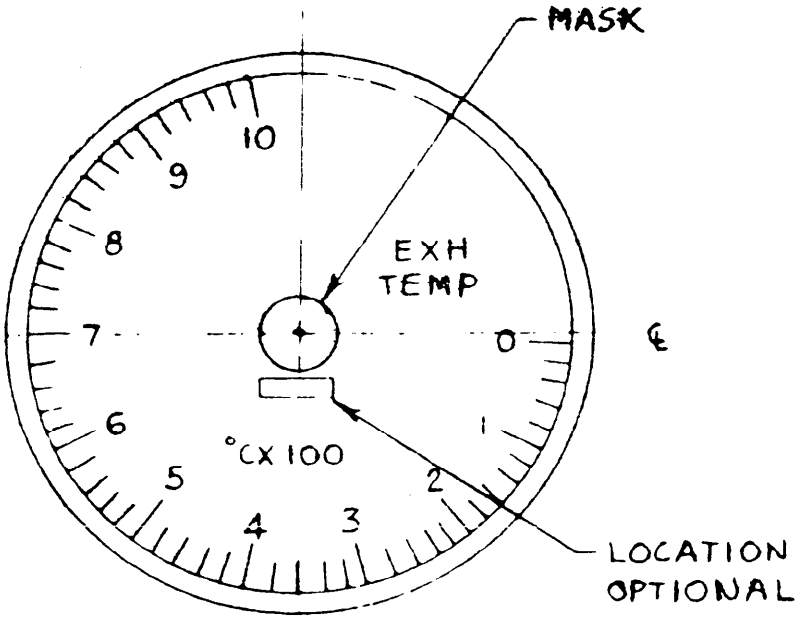


Figure 2. Dial

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other metal material that will permit compliance with the lighting and other requirements specified herein.

3.3.3 Scale Arc. The scale arc of the indicator shall be not less than 250°. The mechanism shall not be geared to obtain the specified scale arc.

3.3.3.1 Scale Distribution. The scale of the indicator shall be as linear as is practicable and shall not be constricted or expanded by more than 1 angular degree between any two adjacent 20°C scale graduations within the range of 200° to 900°C. The remainder of the scale shall not be constricted or expanded by more than 1.5 angular degrees between any 2 adjacent scale graduations. The scales are based on a 5 angular degree nominal spacing between 20°C scale graduations.

*3.3.4 Pointer. The width of the pointer 0.50 ± 0.063 inch from the tip shall be 0.078 ± 0.016 inch and shall taper to a width of 0.031 ± 0.01 inch at the tip. The pointer tip shall extend into the longest scale graduation a distance of 1/16 ± 1/64 inch.

3.3.4.1 Pointer overswing. The pointer overswing shall be not less than 5 angular degrees deflection beyond each end of the scale.

*3.3.5 Lighting. The dial and pointer shall be integrally lighted in accordance with MIL-L-25467 except that the MS24367 lamp shall be used. The color brightness for the white color of the dial and numerals shall be 2.4 ± 1.2 foot-lamberts, 2.6 ± 1.3 foot-lamberts for the pointer, and 0.12 ± 0.06 foot-lamberts for the black background.

*3.3.6 Dial to cover glass distance. The distance between the indicator dial and the cover glass shall not exceed 0.188 inch.

3.4 Weight. The weight of the complete indicator shall not exceed 1.0 pound.

*3.5 Electrical receptacle. An M25516/17-03 receptacle and mating plug conforming to MIL-C-25516-17D shall be provided.

*3.5.1 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

3.6 MARKINGS

3.6.1 Dimensions and color. The dimensions and color of the markings shall be as follows. The colors shall be in accordance with FED-STD-595.

* Markings	Height or Length ± 0.005	Width of Line or Graduation ± 0.005	Color
Numbers	0.156	0.025	37875
All Major Graduations	0.156	0.025	37875
All Minor Graduations	0.093	0.015	37875
"EXH TEMP"	0.125	0.025	37875
"°C X 100"	0.093	0.015	37875
EHU-4/A	0.062	0.010	37038
Pointer	-----	-----	37875
Background of dial	-----	-----	37038
Mask	-----	-----	37038

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3.6.2 Case back. The back of the indicator case shall be distinctly marked so as to indicate "plus" for the chromel-terminal stud and "minus" for the alumel-terminal stud. The statement, "Use 8 ohm special C-A leads," shall be plainly marked on the rear of the case.

3.7 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

*4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Test tolerances. The tolerances specified in Table I, not enclosed in parentheses, are applicable only when the indicators are subjected to all the tests in the sequence specified herein. The tolerances specified in Table I, enclosed in parentheses, are applicable only when the indicators are subjected to a specific test independent of sequence.

4.3 ACCEPTANCE TESTS

4.3.1 INDIVIDUAL TESTS

4.3.1.1 Scale error at room temperature. The scale errors at the test points specified in Table I shall not exceed the tolerances specified in Table I. When this test is conducted, the pointer shall be set to read 700°C when the millivoltage for 700°C as specified in Table I of MIL-I-7071 for the ambient room temperature, is applied to the terminals of the indicator through the appropriate resistance in series with the indicator. If more accurate over-all indicator accuracy can be obtained by setting the pointer to 700° ± 5°C with the specified voltage for 700°C, the pointer shall be so positioned. Unless otherwise specified, the indicator shall be internally lighted during all tests. The voltage applied to the lamps shall be 4.5 ± 0.5V ac or dc.

4.3.1.2 Position errors. This test may be conducted at any point on the scale between 200° and 900°C by applying test voltage. The maximum change in indication shall not exceed 25°C. This test shall be conducted at a major or minor scale graduation.

4.3.1.3 Friction. The maximum difference in reading, before and after tapping, shall not exceed 30°C.

4.3.1.4 Zero adjustment. Power shall be applied to produce a 700°C reading. When the indicator zero adjustment is operated throughout its range, the resulting pointer motion shall be not less than 40°C above and below the 700°C reading. The total motion of the pointer shall not exceed 200°C. The pointer shall be reset to the initial reading with the initial power applied. No adjustment shall be made to the indicator for any of the remaining tests of this specification.

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TABLE I

Exhaust Temperature Test Tolerance

Test point	Scale error at room temperature	Scale error at low temperature	Scale error at high temperature	Scale error at room temperature after vibration failure	Extreme temperature exposure		
					Scale error at low temperature	Scale error at room temperature after low temperature	Scale error at room temperature after high temperature
°C	Tolerance ±°C	Tolerance ±°C	Tolerance ±°C	Tolerance ±°C	Tolerance ±°C	Tolerance ±°C	Tolerance ±°C
0	40 (40)	50 (50)	50 (50)	50 (50)	60 (50)	50 (40)	50 (40)
200	30 (30)	40 (40)	40 (40)	40 (40)	50 (40)	40 (35)	40 (35)
400	20 (20)	35 (35)	30 (30)	35 (30)	45 (40)	40 (30)	40 (30)
500	20 (20)	35 (35)	30 (30)	35 (30)	45 (40)	40 (30)	40 (30)
600	10 (10)	25 (25)	20 (20)	25 (20)	40 (30)	30 (20)	30 (20)
700	5 (5)	20 (20)	15 (15)	20 (15)	35 (25)	25 (15)	25 (15)
800	10 (10)	25 (25)	20 (20)	25 (20)	40 (30)	30 (20)	30 (20)
900	20 (20)	35 (35)	30 (30)	35 (30)	45 (40)	40 (35)	40 (25)
1,000	30 (30)	40 (40)	40 (40)	40 (35)	45 (40)	40 (35)	40 (35)

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4.3.1.5 Insulation leakage. The insulation resistance shall be a minimum of 20 megohms.

4.3.2 SAMPLING PLAN A

4.3.2.1 Scale error at low temperature. The scale errors shall not exceed the tolerances specified in Table I. If there is any tendency for the pointer to stick to the pointer stops, a light tap on the indicator shall be sufficient to free the pointer.

4.3.2.2 Scale error at high temperature. The scale errors shall not exceed the tolerances specified in Table I. If there is any tendency for the pointer to stick to the pointer stops, a light tap on the indicator shall be sufficient to free the pointer.

4.3.2.3 Magnetic effect. When the indicator is subjected to the magnetic effect test, it shall not cause the magnetic compass to deflect more than 5°.

4.3.2.4 Vibration error. The maximum amplitude of pointer oscillation throughout the frequency range shall not exceed a range of 25°C within the temperature range of 400° to 900°C and shall not exceed 30°C for the remainder of the scale. The maximum difference between the original position of the pointer before vibration and the position of the pointer during vibration shall not exceed 15°C between 400° and 900°C and shall not exceed 25°C for the remainder of the scale. Pointer oscillation shall not be considered as pointer variation. The diameter of the circular motion of vibration shall be 0.009 to 0.011 inch.

4.3.2.5 Vibration failure. The scale errors at room temperture, after vibration, shall not exceed the tolerance specified in Table I.

4.3.2.6 Resistance. The resistance of the indicator shall be 42.50 $\pm$ 2.50 ohms.

4.3.3 SAMPLING PLAN B

4.3.3.1 Cold junction compensator lag. The indicator shall be connected to a chromel-alumel thermocouple (8 ohms total external-circuit resistance), immersed in an ice bath of 0°C, and a millivoltage of constant value applied in series with the indicator and thermocouple to provide readings of 700°C. The indicator shall then be subjected to the conditions specified in MIL-1-7071 for increasing and decreasing temperatures. The indication of the indicator throughout the test shall at no time vary more than 30°C from the indication obtained at room temperature.

4.3.3.2 Extreme low temperature exposure. The scale errors at low temperature shall not exceed the tolerances specified in Table I. The scale errors at room temperature, after low temperature exposure, shall not exceed the tolerance specified in Table I.

4.3.3.3 High temperature exposure. The scale errors at room temperature after high temperature exposure shall not exceed the tolerances specified in Table I.

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4.3.4 Thermal shock. Excess peeling of the paint and fogging of the cover glass shall be causes for rejection.

*5. PACKAGING

*5.1 Packaging and packing. Indicators shall be individually packaged and packed in accordance with MIL-STD-794. Preservation shall be in accordance with method II of MIL-P-116.

5.2 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Indicator, Temperature, Thermocouple EHU-4/A

*5.2.1 Reinspection dates. Reinspection marking shall be added in accordance with the instruction of the procuring activity.

6. NOTES

6.1 Intended use. The Type EHU-4/A indicator covered by this specification is intended for use in indicating exhaust temperatures of jet engines.

*6.1.1 The temperature indicator can be used on instrument panels requiring case mounting, in accordance with MS33639, by the application of either an instrument mounting bezel in accordance with MIL-B-6836 or a mounting clamp in accordance with MS28042-1.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Level of packaging and packing desired.

c. Whether sampling plan B tests are to be conducted.

*6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is San Antonio ALC/MMIREC, Kelly AFB, Texas 78241 and information pertaining to qualification of products may be obtained from that activity.

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*6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-I-25852 Rev B Indicator, Temperature, Thermocouple EHU-4/A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   [ ] YES [ ] NO *(If "Yes", in what way?)*

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